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Diagnosis of Parkinson's Diseases with MDVP Features Using Machine Learning Approaches

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***Abstract*—Parkinson's disease is a neurodegenerative disorder that poses a significant global health challenge. Its prevalence has prompted the urgent need for fast and immediate diagnosis to enable timely intervention and treatment planning. Parkinson's disease causes both motor and non-motor symptoms necessitating a comprehensive approach to prediction. This research explores the intricacies of Parkinson's disease prediction, highlighting the importance of multidimensional data analysis. Acknowledging the evolving landscape of medical research, the study advocates for a comprehensive approach to enhance the early detection of Parkinson's disease, provides significant insights on the possibilities of machine learning techniques for improved patient outcomes, particularly focusing on the distinctive features offered by MDVP (Multidimensional Voice Program). Including MDVP features enhances the predictive capabilities of the models, offering a novel perspective for accurate early detection and applying machine learning models like AdaBoost, Multilayer Perceptron, Decision Tree, Random Forest, Logistic Regression, K-nearest Neighbors, Support Vector Machine (SVM), Gradient Boosting, & Naïve Bayes. This research explores the intricacies of Parkinson's disease prediction. Results highlight the remarkable performance of the Support Vector Machine, achieving an accuracy of 98.98% & Gradient Boosting, 97.28% accuracy with the feature selection technique XGBoost and SelectKBest underscoring the significance of MDVP features in refining the predictive accuracy. This study leads to continuous efforts for the improvement of diagnostic accuracy and prognosis for Parkinson's disease, emphasizing the importance of integrating MDVP features within the machine learning framework.**

Keywords— Machine learning, Parkinson's disease, Multidimensional Voice Program, Support Vector Machine

I. INTRODUCTION

Parkinson's disease (PD) is a degenerative neurological condition affecting the brain. The underlying etiology of Parkinson's disease is uncertain. But, the study indicates that the combination of biological and genetic factors plays a vital part in the progression of Parkinson's disease. [1]. For a broad understanding, Parkinson's disease is characterized by the progressive degeneration of nerve cells in various areas of the brain, resulting in a variety of disorders. One significant aspect involves the malfunction of substantia nigra cells are liable for generating dopamine, a neurotransmitter crucial for smooth movement coordination. The intricate communication network within the brain heavily relies on dopamine, acting as a messenger for transmitting signals. The decline in these cells results in movement challenges for patients. Beyond the well-known motor symptoms such as tremors and slowness of movement [2]. Parkinson's encompasses motor symptoms, affecting aspects like mood, sleep, and cognition. Effective Parkinson's disease therapy requires treating both motor and non-motor features.

This work considers features from a voice sample retrieved by a computer program called the Multi-Dimensional Speech Program (MDVP). Using the collected data, a model for detecting early Parkinson's can be developed. Here, we attempt to apply several machine-learning techniques to boost the model's accuracy. While also playing a significant part in the early diagnosis of Parkinson's Disease, allowing neuroprotective medications to be initiated at the appropriate time. Currently, there is no definitive tool for Parkinson's disease diagnosis [3]. However, multiple signs and tests for diagnosis are employed in conjunction. Scientists have examined many signs to diagnose Parkinson's disease earlier and stop its development. At present, PD drugs only alleviate symptoms and do not slow or stop the disease's progression. Several strategies have been developed to assist in identifying Parkinson's disease using different indicators, such as speech data [4,5], gait patterns [6], tremor data [7], smell identification data [8], and spontaneous cardiovascular oscillations [9].

Researchers applied several traits and data to forecast Parkinson's disease. Rehman et al. [10] utilized a biomedical voice analysis tool, MDVP, for human voice analysis and achieved an accuracy of 95% using LSTM+GRU deep learning model. Thanoun et al. [11] presented a new approach to predict Parkinson's disease. Because of a disparity in their dataset, they used SMOTE to resolve the imbalance within the dataset. They then used the balanced dataset to train an Extra Trees classifier with k-fold cross-validation. Despite the unbalanced data distribution, their algorithm correctly classified Parkinson's disease as 96.52%. Wang, Wu, et al. [12] developed a model using deep learning that recognizes PD patients and normal persons based on premotor markers (such as olfactory loss, sleep behavior disorder (RBD), and rapid eye movement (REM) sleep). Their designed deep learning model achieved 96.45% accuracy on average. Govindu et al. [13] worked with biomedical voice measurement, MDVP, of 31 people where 23 were PD. They

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trained their model using several machine learning algorithms which yielded an accuracy of 91.83% and precision is 95% & a sensitivity of 90.55%. Mathur et al. [14] use several machine learning algorithms to predict Parkinson's disease and get 91.28% accuracy by combining K-nearest Neighbors and Multilayer Perceptron. Abba, Sani, et al. [15] examined and assessed a range of supervised machine learning algorithms to identify people who may have Parkinson's disease, they worked with various aspects of voice by MDVP analysis & got the accuracy of 98% by gradient boosting model. Inspired by this research, in the current research, we aim to extend the study by improving accuracy through the use of sophisticated machine-learning algorithms and preprocessing. The primary contributions of our work are as follows:

- In data preprocessing, we used to enhance the quality and usability of raw data for analysis by applying various techniques such as label encoding, feature selection, and imbalanced data handling.
- We encode data and transform categorical data to numerical format, facilitating the compatibility of such variables with machine learning algorithms.
- Random Over-sampling is used to address class imbalance by increasing the representation of minority classes, thereby increasing the performance and equity of machine learning models.

II. METHODOLOGY

We collected the dataset for this study from UCI machine learning repositories [16]. Initial data is made up of several biomedical voice metrics. Firstly, the system applied several data preprocessing approaches including data encoding, feature selection & random oversampling. For feature selection, we used three feature selection methods, selectKBest, XGBoost, and PCA. After that split the dataset into 80% & 20% for train and test models respectively. Several supervised machine learning classifiers were employed to identify the top-performing model. The models are subjected to training data to learn from their pattern and appraised based on testing data to determine the accuracy, precision, sensitivity, specificity, and f-measure. Moreover, the machine learning model also clarified the organizational structure of the system and the interactions between its many components. "Fig. 1" illustrates the proposed model's overall architecture.

A. Dataset Overview

The datasets were collected from UCI machine learning repositories [16], where a number of biological speech metrics from 31 people, 23 of whom have Parkinson's disease (PD). Each row in the dataset represents one of the 195 voice recordings from the people ("name" column), with each column representing a distinct voice measurement. The "status" column is set to 0 for healthy and 1 for PD and serves as the primary means of distinguishing between persons without Parkinson's disease (PD) and those with it. We used various machine learning models like AdaBoost, Multilayer Perceptron, Decision Tree, Random Forest, Logistic Regression, K-nearest Neighbors, Support Vector Machine (SVM), Gradient Boosting, & Naïve Bayes to detect Parkinson's Diseases.

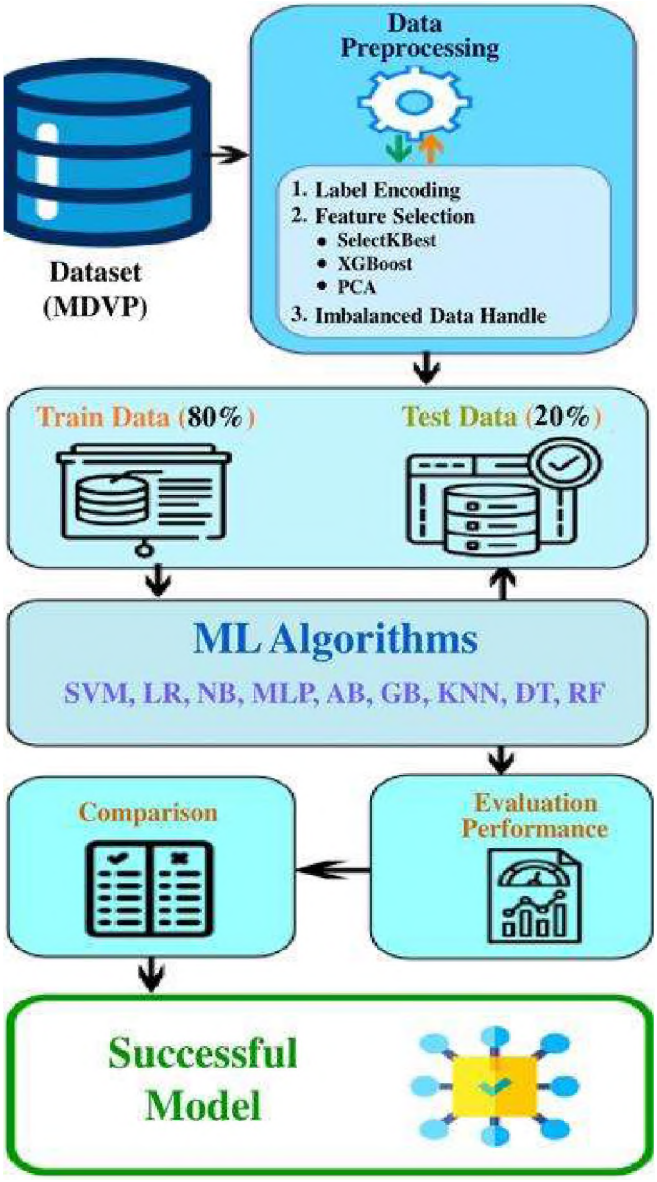


Fig. 1. Proposed Methodology of our system

B. Data Preprocessing

Preprocessing is an essential stage in data analysis and machine learning which involves cleaning, organizing, and converting initial information into a format required for study or model training. The primary goals of preprocessing are to handle missing data, standardize formats, and extract relevant features, ultimately improving the overall data quality.

Label Encoding

Label encoding is a data preprocessing technique that converts category data to numerical form. Label encoding generates a unique number for every category or label in a categorical parameter.

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Feature Selection

The phonation dataset provides a total of 22 vocal features provided by the dataset but in feature selection 16 of the features are taken from those details outlined in Table I.

TABLE I. ATTRIBUTES OF DATASET

No.	Attribute	No.	Attribute
1	MDVP:Fo(Hz)	9	MDVP:APQ
2	MDVPJ:Jitter(%)	10	HNR
3	MDVP:Jitter(Abs)	11	Status
4	MDVP:RAP	12	RPDE
5	Jitter:DDP	13	Spread1
6	MDVP:Shimmer(dB)	14	Spread2
7	Shimmer:APQ3	15	D2
8	Shimmer:APQ5	16	PPE

1. Vocal fundamental frequencies
   - Multidimensional Voice Program: Fo (Hz)
2. Variance in fundamental frequencies
   - MDVP: Jitter (%), MDVP: Jitter (Abs) - Indicators of nervousness, measured in percentage and absolute milliseconds, respectively, highlighting abnormalities in vocal cord vibration.
   - MDVP: RAP (%) - RAP (Relative Average Perturbation) is calculated by determining the variance among a period and its average, considering its two neighbors, and then dividing it by the whole duration.
   - Jitter: DDP (%) - To calculate Jitter: DDP, multiply the value of RAP by 3.
3. Variations in amplitude
   - MDVP: Shimmer (dB) - The "dB" in MDVP, Shimmer (dB) indicates that the shimmer values are expressed in decibels, which is a unit of measurement for the intensity or loudness of a sound.
   - MDVP: APQ - Amplitude Perturbation Quotient measures the variation in fundamental frequencies.
4. Noise-to-harmonics ratio (NHR) measures the balance between noise and tonal components in the voice.
   - NHR: NHR - Noise-to-Harmonics Ratio.
   - HNR - Harmonics-to-Noise Ratio.
   - Status - Healthy patients are denoted by 0, and PD patients by 1.
   - Nonlinear dynamical complexity measures.
   - RPDE - Recurrence Period Density Entropy determines the periodicity of a signal.
   - D2 - Also known as correlation dimension.
   - spread1, spread2, PPE - Three nonlinear measures of fundamental frequency variation

C. Imbalanced Data Handle

Handling imbalanced data is a common challenge in machine learning, especially in binary classification problems where one class significantly outnumbers the other.

"Fig. 2' visualizes the unbalanced data before feature optimization. Since imbalanced data can lead to biased models that perform poorly on the minority class, therefore we have applied Random Oversampling on the dataset to balance data, as ROS employed in the field of imbalanced data classification [17]. "Fig. 3" depicts the visualization of balanced data after the application of Random Oversampling.

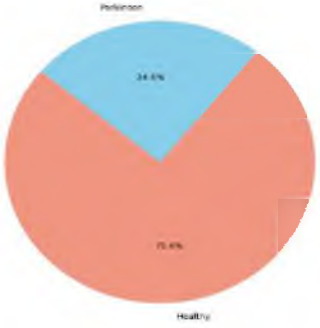


Fig. 2. Distribution of Healthy and Unhealthy Patients before ROS

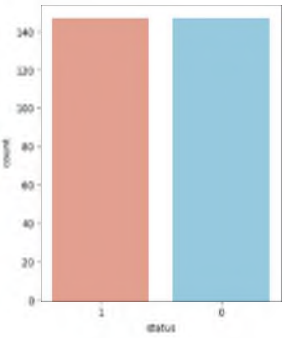


Fig. 3. Impact of ROS on Class Distribution

2.4 Exploratory Data Analysis

We examined the raw data using Boxplot in "Fig. 4." to glean insights into the features. To comprehend the range, histograms were generated for each feature individually. Presented here are the histograms for two specific parameters.

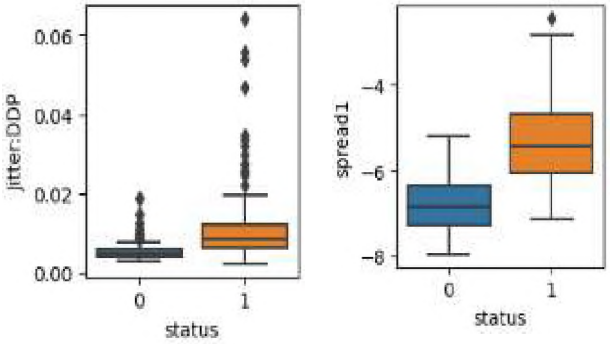


Fig. 4. Sample Dataset Boxplot

To comprehend the connections between the variables in a dataset, we also look at the correlation matrix. The correlation matrix aids in understanding the relationships among the features, encompassing positive, negative, and zero correlations. Positive correlation means parameters change together, whilst negative correlation implies opposing value changes; an increase in one corresponds to a decrease in the other and vice versa. A zero correlation implies complete independence between variables. Upon focusing solely on the status column of the data, notable high negative correlations with the output category are observed for HNR and fundamental frequencies. It is noteworthy that these features significantly affect the status column, helping to the identification of Parkinson's disease. Additionally, this extensive investigation adds to the knowledge of the complex links between distinct vocal qualities and the existence of Parkinson's disease. The model was constructed using data from the analysis of 16 parameters of the Multidimensional Voice Program (MDVP).

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D. Machine Learning Classifiers

Support Vector Machine (SVM):

Support Vector Machine (SVM) is a machine learning technique used for both regression and classification tasks [18]. It aims to identify a hyperplane that best distinguishes distinct classes in the feature space. The linear SVM decision function can be stated as:

$$f(x) = w \cdot x + b \quad (1)$$

Here, f(x) is the decision function, w represents the weights assigned to features, The bias component is b, whereas x represents the input feature vector.

If f(x) is positive, the data point goes to one class, and if negative, it belongs to the other.

Gradient Boosting (GB):

Gradient Boosting is a method of ensemble learning that creates a predictive model in stages by integrating the predictions of numerous weak learners [19]. It seeks to reduce the mistakes of prior models by fitting new models to the residuals. The final model is the weighted total of the weak learners.

The general formula for the additive model in the case of gradient boosting is:

$$F(x) = \sum_{m=1}^{M} \beta m hm(x) \quad (2)$$

Where, $F(x)$ represents the final model, M is the number of weak learners (trees), βm is the weight that is assigned to the m-th weak learner, and $hm(x)$ is the prediction of the m-th weak learner.

III. Results

Various ML classifiers, including AB, MLP, DT, RF, LR, KNN, SVM, GB, and NB, were used in this research, with SVM and GB outperforming the feature selection techniques SelectKBest, XGBoost, and PCA with 98.98% and 97.28%, respectively.

A. Confusion Matrix

To get a better idea of our model's performance, we employed a confusion matrix [20] to illustrate our results. For our objective of achieving a binary classifier, we utilized a 2x2 confusion matrix. Here, True positives are denoted by TP, true negatives by TN, false positives by FP, and false negatives by FN [21]. Equations 3 to 7 help to calculate Accuracy, Sensitivity, Specificity, Precision, and F-Measure respectively [22].

$$Accuracy = \frac{(TP+TN)}{(TP+TN+FP+FN)} \quad (3)$$

$$\text{Sensitivity} = \frac{TP}{(TP+FN)} \quad (4)$$

$$\text{Specificity} = \frac{TN}{TN+FP} \quad (5)$$

$$\text{Precision} = \frac{(TP)}{(TP+FP)} \quad (6)$$

$$\text{F} - \text{Measure} = \frac{TP}{TP+0.5(FP+FN)} \quad (7)$$

From the machine learning models, the Support Vector Machine gives the best result of accuracy of 98.98%. The high accuracy of SVM classifiers underscores its effectiveness in accurately classifying instances in the dataset.

Table II. shows the model evaluation for 4 performance indicators: Sensitivity, Specificity, Precision & F-Measure.

TABLE II. PERFORMANCE EVALUATION OF DIFFERENT ML MODELS

Model	Sensitivity	Specificity	Precision	F-Measure
SVM	98%	100%	100 %	98.99%
LR	82.58%	86.33%	87.07 %	84.77%
GB	95.42%	99.29%	99.32 %	97.33%
KNN	93.59%	99.28%	99.32 %	96.37%
DT	87.12%	96.18%	96.60 %	91.61%
NB	74.25%	81.89%	84.35 %	78.98%
RF	90.63%	98.51%	98.64 %	94.46%
AB	89.57%	99.24%	99.32 %	94.19%
MP	87.35%	98.44%	98.64 %	92.65%

Model assessment involves examining the performance and efficacy of predictive models using a variety of indicators and methodologies to assure their dependability and applicability.

The accuracy of each ML classifier is shown in "Fig 5", highlighting the performance across various models and providing valuable insights into their comparative effectiveness.

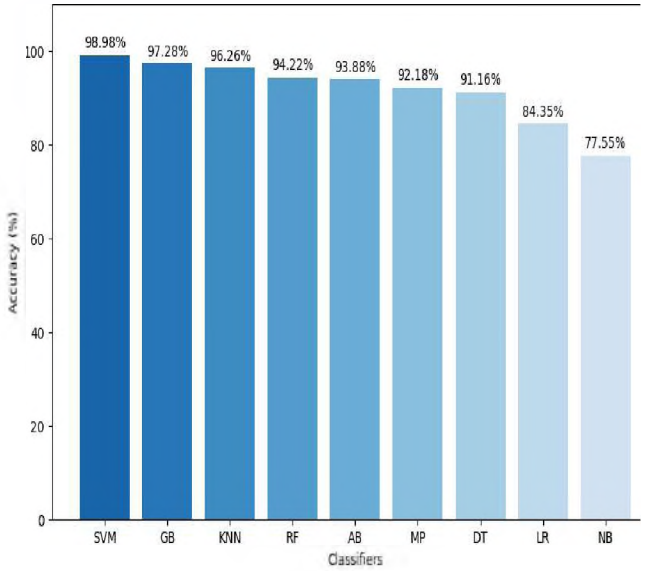


Fig. 5. Comparison of accuracy of different ML classifiers

B. ROC and AUC

The ROC curve is an illustration for evaluating the results of binary models for classification [23]. It shows the True Positive Rate vs the False Positive Rate at different judgment criteria. The curve illustrates the compromise between

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sensitivity and specificity, enabling the selection of an ideal threshold depending on the application's unique requirements. The Area Under the Curve (AUC) is a quantitative statistic obtained from the ROC curve [24]. It assesses a classification model's overall discriminative power by evaluating the area beneath the ROC curve. AUC values vary from 0 to 1, with 0.5 suggesting that a model perfoms similarly to random chance and 1.0 signifying perfect discrimination. "Fig. 6" shows the ROC curve and AUC values for each of the nine machine-learning models.

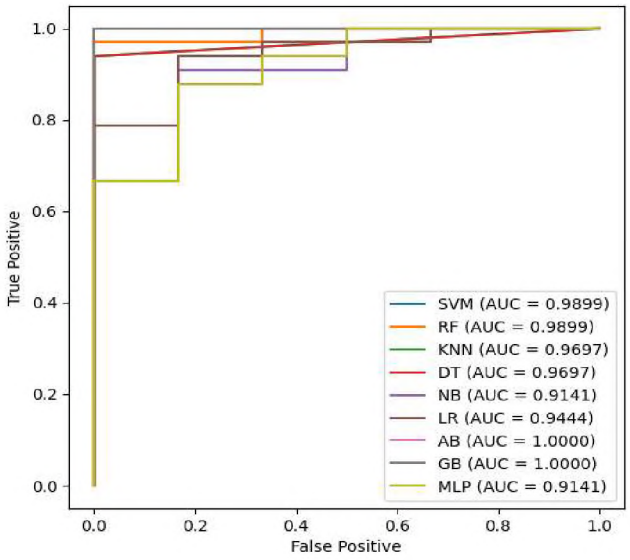


Fig. 6. ML Model's evaluation based on AUC ROC curves

IV. Discussion

In an effort to increase the accuracy of Parkinson's disease prediction, our study employed various machine learning classifiers on a dataset comprising MDVP features. The classification models utilized AdaBoost, Multilayer Perceptron, Decision Tree, Random Forest, Logistic Regression, K-nearest Neighbors, Support Vector Machine (SVM), Gradient Boosting, & Naïve Bayes. Among the classifiers, SVM achieved the highest accuracy of 98.98%, demonstrating its ability to discriminate between healthy people and those harmed by Parkinson's disease. These promising results contribute to the growing body of research in medical diagnostics and also highlight the significance of leveraging machine learning methodologies for improved healthcare outcomes. Further exploration and validation of these models on diverse datasets may provide valuable insights for the development of robust and reliable diagnostic tools for Parkinson's disease. In comparison with previous research work we have mentioned earlier, Rehman et al. [10] utilized a biomedical voice analysis tool to predict Parkinson's disease and achieved an accuracy of 95% using LSTM+GRU deep learning model. Thanoun et al. [11] used SMOTE to resolve dataset imbalance, then trained an Extra Trees classifier with k-fold cross-validation, accurately classifying Parkinson's disease at 96.52%. Wang, Wu, et al. [12] developed a deep learning model that accurately differentiates PD patients from normal individuals using premotor markers, achieving an average accuracy of 96.45%. Govindu et al. [13] work with biomedical voice measurement, MDVP. They use several machine learning models which obtained an accuracy of 91.83%. Mathur et al. [14] use several machine learning algorithms to predict Parkinson's disease and get 91.28% accuracy by combining K-nearest Neighbors and Multilayer Perceptron. Abba, Sani, et al. [15] worked with various aspects of voice by MDVP analysis & got an accuracy of 97.5%. On the other hand, we gained better accuracy, with the majority of the classifiers operating satisfactorily. Additionally, the Support Vector Machine performed the best, achieving an accuracy of 98.98%.

Table III shows the result of our proposed model compared to previous research performance.

TABLE III. **Comparative Analysis of Existing Method and Our Proposed Method**

Studies	Model	Accuracy	Precision	F-Measure
Rehman et al. [10]	LSTM+GRU	95%	94%	95%
Thanoun et al.[11]	Random Forest	96.52%	96%	95%
Wang, Wu, et al. [12]	Deep Learning Model	96.45%	97.53%	97.42%
Govindu et al. [13]	K-nearest Neighbour	91.83%	95%	95%
Mathur et al. [14]	KNN+MLP	91.28%	87.30%	88.80%
Abba, Sani, et al. [15]	Extreme Gradient Boosting	98.00%	97.50%	98.7%
Present Model	**Support Vector Machine**	**98.98%**	**100%**	**99.99%**

V. Conclusions

A high-efficiency model was developed for the diagnosis of Parkinson's disease. This comparative analysis showcased the utility of a voice dataset from Parkinson's disease patients, employing Support Vector Machines. The techniques utilized were benchmarked against previous machine learning models. To further enhance the model's capabilities, future iterations may incorporate an additional dataset involving the analysis of spiral drawings from Parkinson's disease patients. This integration aims to bolster the scalability of the model. Furthermore, a prospective avenue for improvement involves implementing machine learning, to yield an enhanced and versatile model. Additionally, ongoing efforts could focus on refining feature selection to enhance the model's applicability to the wider public. Ultimately, this model holds potential applications in healthcare settings, enabling hospitals or clinics to input test results for Parkinson's disease diagnosis.

References

[1] S. Bandres-Ciga, et al., "Genetics of Parkinson's disease: an introspection of its journey towards precision medicine," Neurobiology of disease, vol. 137, 2020, Art. no. 104782.

[2] T. C. Vu, J. G. Nutt, and N. H. G. Holford, "Progression of motor and nonmotor features of Parkinson's disease and their response to

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treatment," British journal of clinical pharmacology, vol. 74, no. 2, pp. 267-283, 2012.

[3] J. Goyal, P. Khandnor, and T. C. Aseri, "Classification, prediction, and monitoring of Parkinson's disease using computer assisted technologies: a comparative analysis," Engineering Applications of Artificial Intelligence, vol. 96, 2020, Art. no. 103955.

[4] I. Ahmed, et al., "Classification of Parkinson disease based on patient's voice signal using machine learning," Intelligent Automation and Soft Computing, vol. 32, no. 2, pp. 705, 2022.

[5] H. Gunduz, "Deep learning-based Parkinson's disease classification using vocal feature sets," IEEE Access, vol. 7, pp. 115540-115551, 2019.

[6] E. Balaji, et al., "Data-driven gait analysis for diagnosis and severity rating of Parkinson's disease," Medical Engineering & Physics, vol. 91, pp. 54-64, 2021.

[7] S. J. K. Prasad, S. M. Kumar, and G. P. Gowda, "Detection and Predictive analytics of Parkinson's disease employing Tremor analysis and Deep Learning Algorithm," NVEO-NATURAL VOLATILES & ESSENTIAL OILS Journal, vol. NVEO, pp. 12320-12340, 2021.

[8] C. C. G. Alonso, et al., "Smell tests to distinguish Parkinson's disease from other neurological disorders: a systematic review and meta-analysis," Expert Review of Neurotherapeutics, vol. 21, no. 3, pp. 365-379, 2021.

[9] G. Valenza, et al., "Assessment of spontaneous cardiovascular oscillations in Parkinson's disease," Biomed. Signal Process. Control, vol. 26, pp. 80-89, Apr. 2016.

[10] A. Rehman, et al., "Parkinson's disease detection using hybrid LSTM-GRU deep learning model," Electronics, vol. 12, no. 13, Art. no. 2856, 2023.

[11] M. Y. Thanoun, M. T. Yaseen, and A. M. Aleesa, "Development of Intelligent Parkinson Disease Detection System Based on Machine Learning Techniques Using Speech Signal," Int. J. Adv. Sci. Eng. Inf. Technol, vol. 11, no. 1, pp. 388-392, 2021.

[12] W. Wang, et al., "Early detection of Parkinson's disease using deep learning and machine learning," IEEE Access, vol. 8, pp. 147635-147646, 2020.

[13] A. Govindu, and S. Palwe, "Early detection of Parkinson's disease using machine learning," Procedia Computer Science, vol. 218, pp. 249-261, 2023.

[14] R. Mathur, V. Pathak, and D. Bandil, "Parkinson disease prediction using machine learning algorithm," Emerging Trends in Expert Applications and Security: Proceedings of ICETEAS 2018, Springer Singapore, 2019.

[15] S. Abba, et al., "Performance Analysis and Evaluation of Supervised Machine Learning Algorithms for the Detection of Parkinson's Disease," 2023.

[16] M. Little, "Parkinson's. UCI machine learning repository," 2008.[Online].Available:https://archive.ics.uci.edu/dataset/174/parkinsons

[17] M. Hayaty, S. Muthmainah, and S. M. Ghufran, "Random and synthetic over-sampling approach to resolve data imbalance in classification," International Journal of Artificial Intelligence Research, vol. 4, no. 2, pp. 86-94, 2020.

[18] D. A. Pisner, and D. M. Schnyer, "Support vector machine," Machine learning, Academic Press, pp. 101-121, 2020.

[19] C. Bentéjac, A. Csörgő, and G. Martínez-Muñoz, "A comparative analysis of gradient boosting algorithms," Artificial Intelligence Review, vol. 54, pp. 1937-1967, 2021.

[20] A. Vanacore, M. S. Pellegrino, and A. Ciardiello, "Fair evaluation of classifier predictive performance based on binary confusion matrix," Computational Statistics, vol. 39, no. 1, pp. 363-383, 2024.

[21] C. Schwenke, and A. G. Schering, "True positives, true negatives, false positives, false negatives," Wiley StatsRef: Statistics Reference Online, 2014.

[22] D. M. Powers, "Evaluation: From Precision, Recall and F-Measure to ROC, Informedness, Markedness & Correlation," Journal of Machine Learning Technologies, vol. 2, no. 1, pp. 37-63, 2011.

[23] T. C. F. Polo, and H. A. Miot, "Use of ROC curves in clinical and experimental studies," Jornal vascular brasileiro, vol. 19, Art. no. e20200186, 2020.

[24] J. R. Turner, "Area under the curve (AUC)," Encyclopedia of Behavioral Medicine, pp. 146-146, 2020.

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